Module 7: Maintain solar photovoltaic system

Unit 7.1: Tools Required for maintenance

Pre-Requisites for Solar Panel Maintenance





Tools required to carry out Solar PV Maintenance are:

Tools	Function
Needle nose pliers	Gripping and manipulating small objects
Linesman pliers	Gripping and cutting wires
Diagonal cutters	Cutting wires and cables
DC soldering iron	Soldering electrical components
Hacksaw	Cutting metal
Compass	Navigation and direction finding
Flashlight	Illumination
Caulking gun	Applying sealant
Utility knife	Cutting and opening materials
Nut drivers 1/4inch and 5/16inch	Turning nuts and bolts
Hammer	Striking and driving objects
Measuring tape (25m)	Measuring distances
Wire strippers	Removing insulation from wires
Cell water filler	Adding water to battery cells
Crimping tool	Crimping connectors to wires

Safety and Measurement Equipment required to carry out Solar PV Maintenance are:

Safety Equipment	Function
Safety goggles	Protect eyes from flying debris, sparks, and chemicals.
Rubber gloves	Protect hands from electrical hazards, chemicals, and sharp objects.
Clamp spreader	Assist in safely separating or spreading clamped objects.

Measuring Equipment	Function
Multimeter	Measures electrical properties like voltage, current, and resistance.
Angle measuring device	Measures angles and degrees.
Combination square	Combines a ruler, level, and try square for multiple measuring tasks.

Documents required to carry out Solar PV Maintenance are:

Documentation	Function
System service logbook	Records maintenance and repair activities performed on a system.
Datasheet & O&M manual	Provides technical specifications and operating instructions for equipment.
Manual	Provides specific instructions and procedures for the task at hand.
Sun Pathfinder	Assists in determining the position of the sun and calculating solar angles.

7.1.1: Solar PV Maintenance

- Regular maintenance of solar PV systems is essential for optimal performance and longevity.
- This includes cleaning panels, inspecting connections, and ensuring proper functioning of components like batteries and inverters.
- By staying proactive, you can maximize energy production and minimize downtime, contributing to a sustainable energy future.



7.1.2: Documentation for Solar Panel Maintenance

- Documentation for solar panel maintenance includes records of cleaning schedules, inspection reports, and any repairs or replacements carried out on components.

- This documentation ensures a systematic approach to maintenance, aids in tracking system performance over time, and facilitates timely interventions to maximize the efficiency and longevity of the solar panel system.



Documentation for Solar Panel Maintenance



Documentation and its Importance





In this video, you will learn about documentation, its need in Solar PV system Maintenance, and the crucial role of commissioning in ensuring optimal system performance.

Documentation and its Importance





Documenting the tools required for solar panel maintenance is essential for safety, efficiency, compliance, and overall effectiveness of the maintenance process.

Documentation and its Importance

For Maintaining Safety:



Accurate Inventory

- Ensures technicians have the correct tools for the job, preventing the use of inappropriate or unsafe equipment.



Training & Familiarization

- Documents can serve as a reference for new technicians, ensuring proper tool usage and safety protocols.



Incident Prevention

- Records of tool usage and potential hazards can help identify and mitigate risks of accidents or injuries.

Documentation and its Importance

For Maintaining Efficiency:



Time Management

• Knowing the required tools before-hand minimizes time spent searching or gathering equipment.



Preventive Maintenance

• Documentation can help schedule tool maintenance and replacement, preventing downtime and ensuring tools are in good working order.



Standardization

• Consistent documentation across maintenance teams promotes efficient workflows and reduces the risk of errors.

Documentation and its Importance

For Maintaining Compliance:



Regulatory Requirements

- Some regulations may mandate specific tools and procedures for solar panel maintenance. Documentation helps demonstrate compliance with these requirements.



Warranty Claims

- Proper documentation of tool usage can be crucial for supporting warranty claims in case of equipment failure.



Traceability

- Records of tool use can help track potential issues or defects back to specific tools or maintenance events.

Logbook

A **logbook** is a record used to document the states, events, or conditions of any equipment over time. It's essentially a journal specifically designed for tracking specific information.

Benefits of a Logbook :



Historical data for performance monitoring



Early detection of potential problems



Improved communication and knowledge transfer

Logbook

Key Information to Include in Logbooks:

 Date and time of maintenance session	Accurate tracking and scheduling.
 Maintenance tasks undertaken	Inspections, repairs, replacements, and adjustments.
 Observations, findings	Abnormalities, issues, or areas requiring further attention.
 Corrective actions implemented	Address observed issues or anomalies, steps taken to resolve problems and ensure optimal functionality.
 Signature for verification	Authorized personnel who confirms the completion of tasks and attests to the accuracy of the recorded information.

Logbook

It is important to maintain a logbook of various tools and equipment that are required for testing of PV systems.





Logbook



First aid kit



Flashlight



Phase sequence meter



Multimeter



Clampon ammeter



Hydrometer



Screwdrivers



Nut drivers



Earth tester



IR Tester



Crimping tool set



Compass



Sun pathfinder



Angle gauge



Battery safety accessories



Battery water filler



Battery maintenance kit

Commissioning a Solar PV System



- By serving as a comprehensive record, the logbook for maintenance equipment plays a pivotal role in ensuring the efficient operation and longevity of Solar PV Systems.
- It establishes a seamless connection between the commissioning phase and ongoing maintenance protocols, facilitating systematic tracking and optimization of system performance.

Commissioning a Solar PV System: A Step-by-Step Guide



1. Preparation

Safety First: Ensure safety as the priority and follow safe work procedures.

PPE: Wear appropriate Personal Protective Equipment (PPE) for the task, including eye protection, gloves, and sturdy footwear.

Weather Conditions: Choose a clear, sunny day for optimal testing and accurate results.



2. Pre-Commissioning Checks

Wiring Security: Double-check all wiring connections for tightness and proper installation.

Isolation: Segment the PV strings by leaving one module interconnection open to prevent accidental arcing.

Power Off: Ensure all circuit breakers and isolators (AC and DC) are in the "off" position, including the inverter.



3. Commissioning

Start at the Array: Begin commissioning at the PV array to identify and rectify any wiring issues before energizing the inverter and potentially causing damage.

Fuses: Verify that all fuses are removed during the commissioning process.

System Checks: Use relevant worksheets and checklists provided in this chapter to conduct thorough inspections and tests.

Unit 7.2: Preventive maintenance of PV System

7.2.1: Solar PV Maintenance Schedule

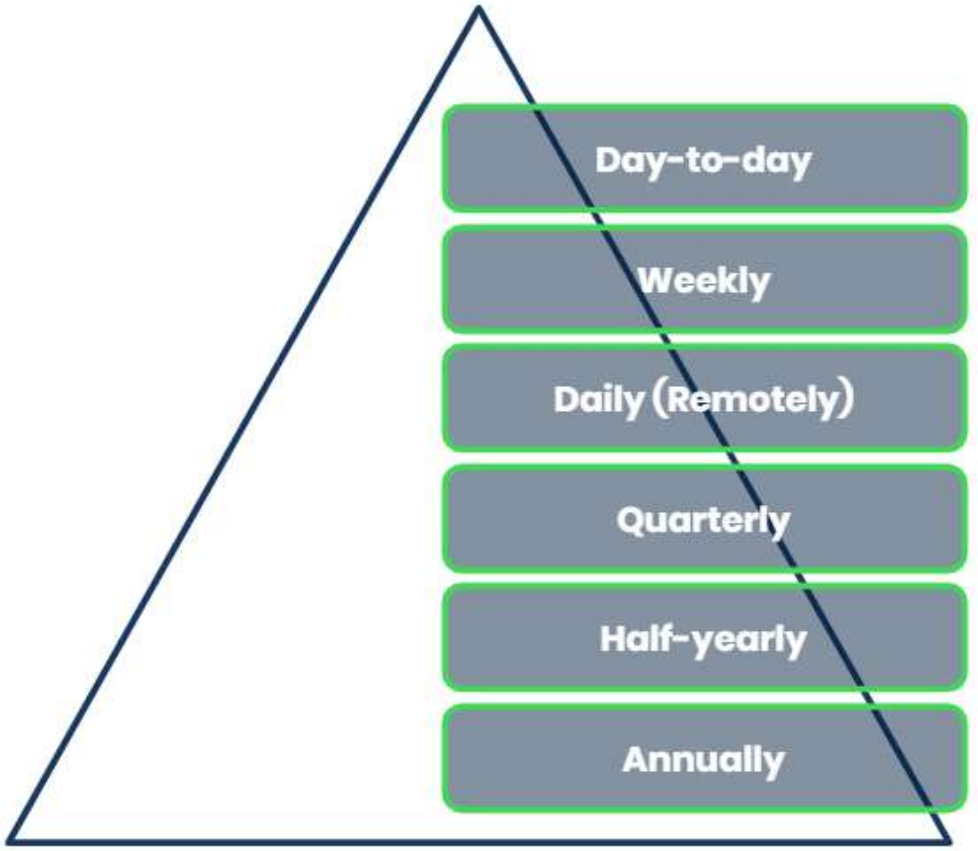
- A maintenance schedule for solar panels outlines regular tasks such as:
  - cleaning,
  - inspection of connections, and
  - monitoring system performance.

- By adhering to a scheduled routine, you ensure the optimal functioning of your solar panel system, maximizing energy production and prolonging its lifespan.

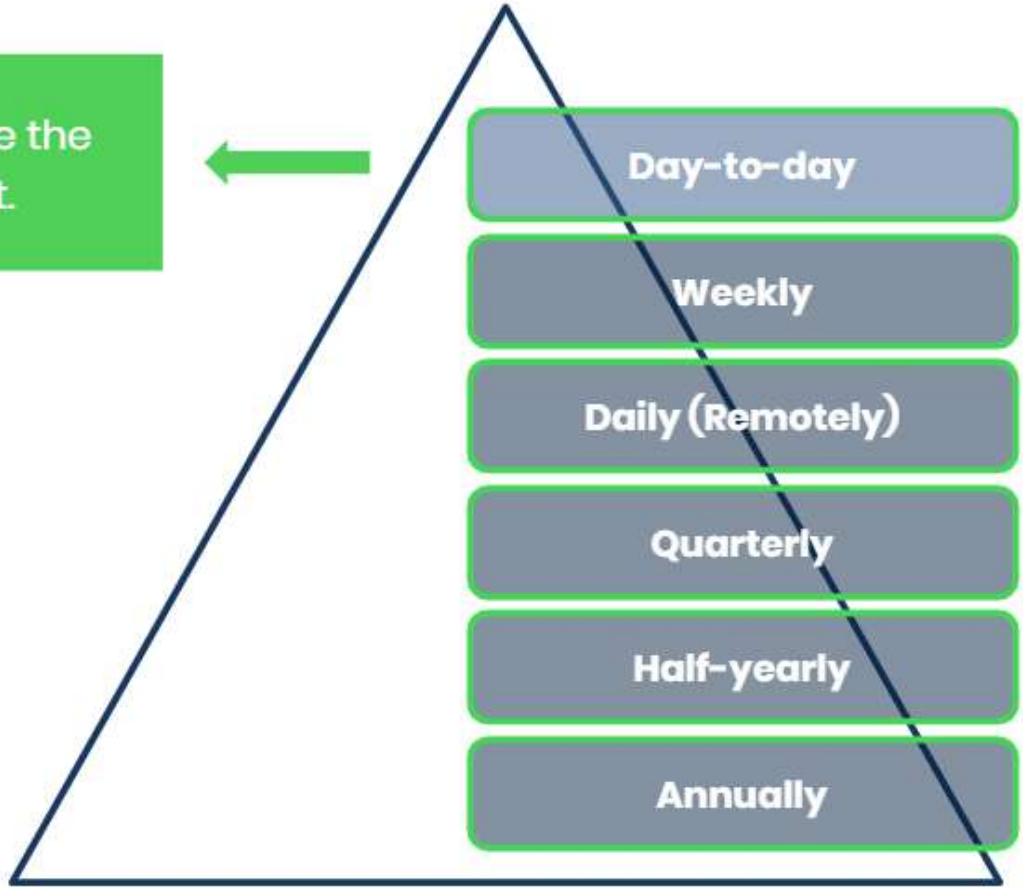


7.2.1: Solar PV Maintenance Schedule

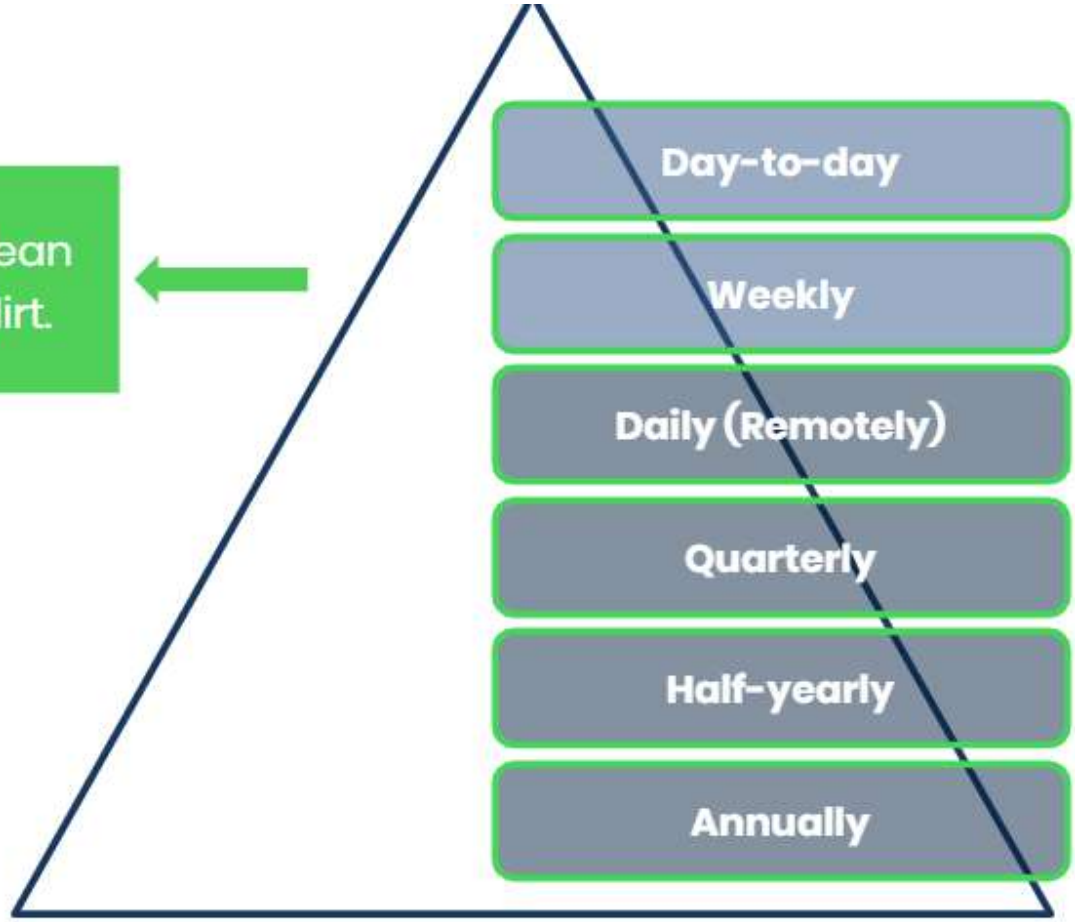
Maintenance tasks that should be performed on a solar panel site are categorized by frequency:



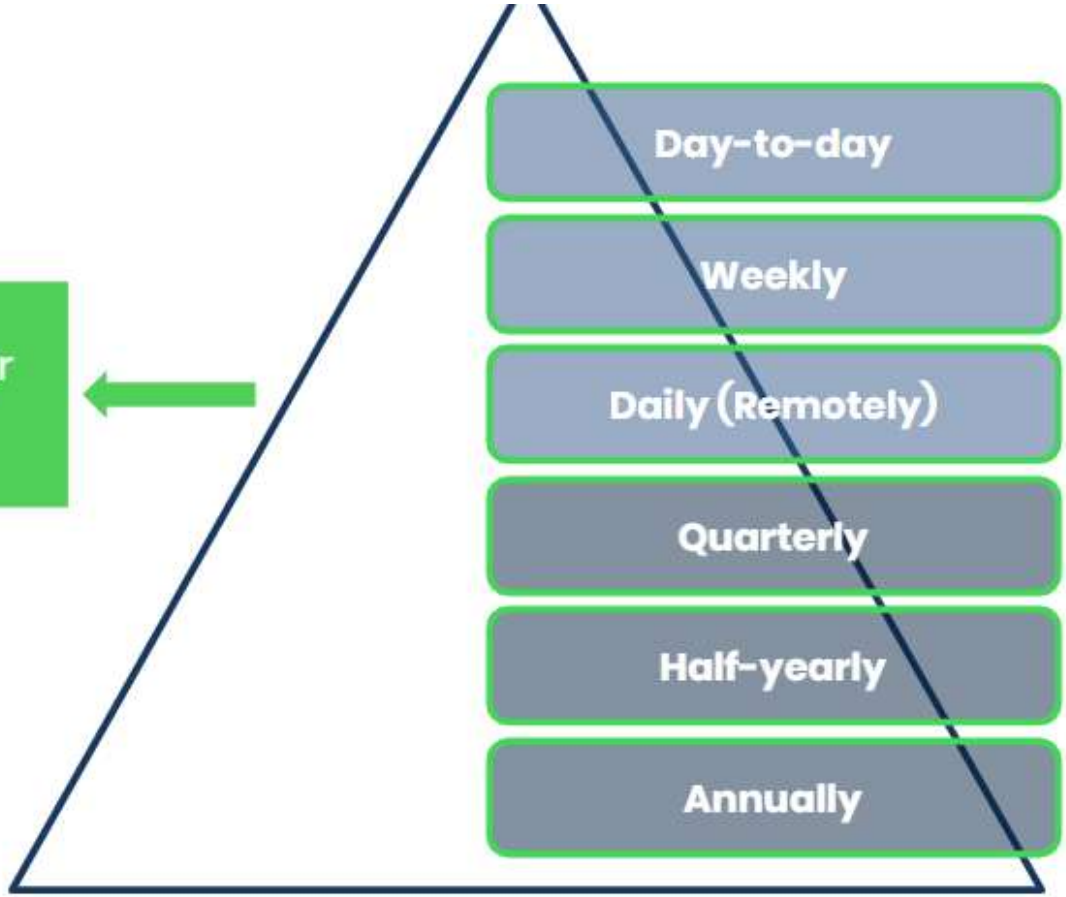
In Day-to-day maintenance, ensure the security of the solar panel plant.



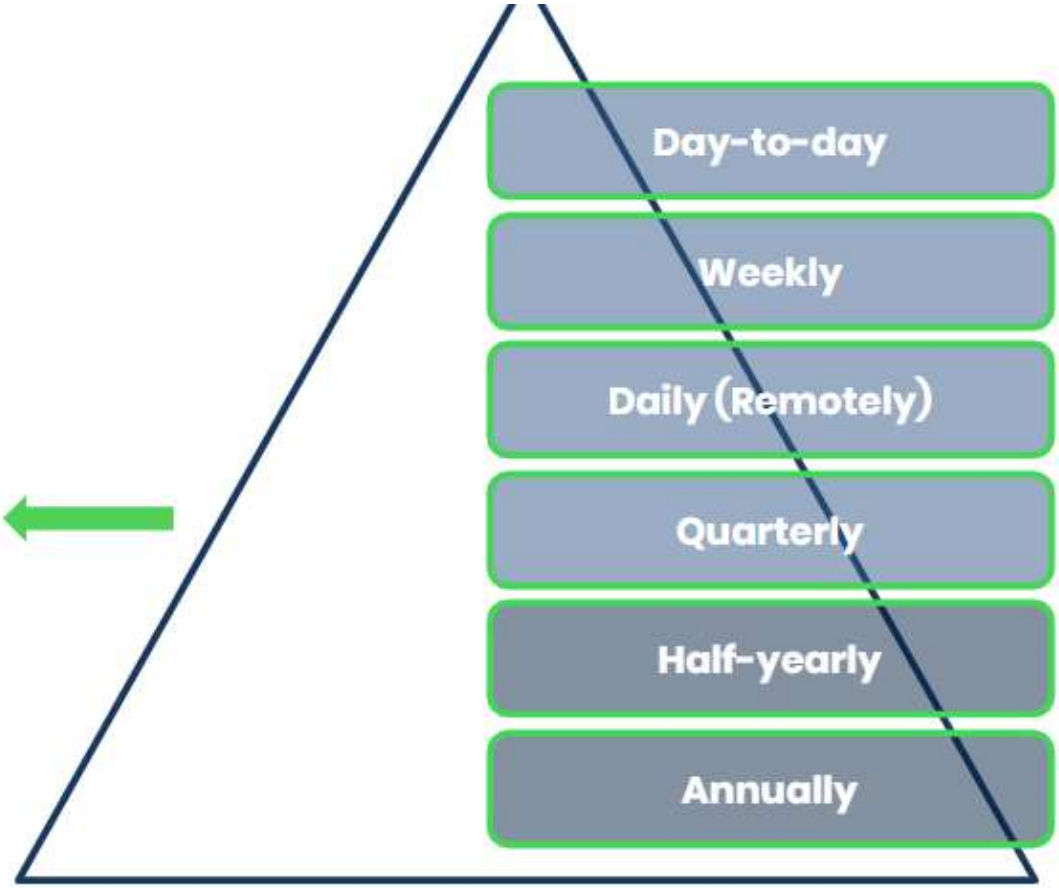
In Weekly maintenance, inspect and clean the PV modules from dust and other dirt.



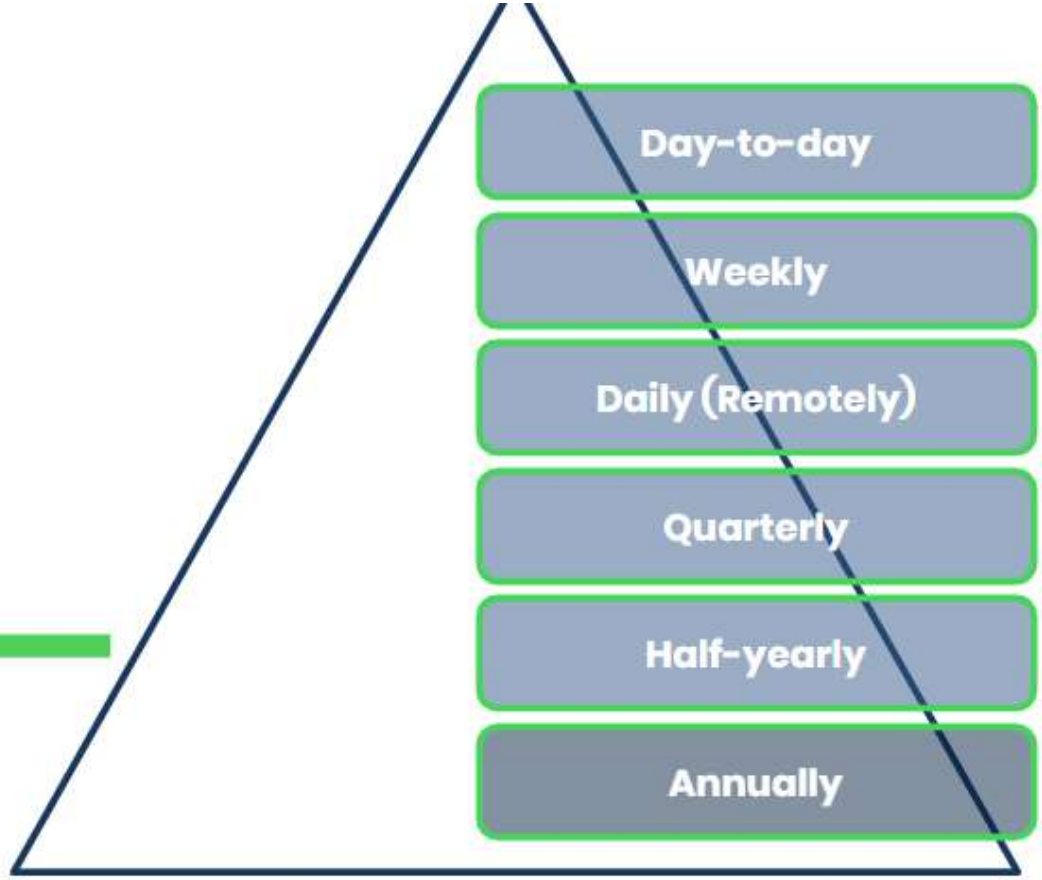
In Daily (Remotely) maintenance, monitor power generation and export.



In Quarterly maintenance, keep the inverters clean to minimize the possibility of dust ingress.

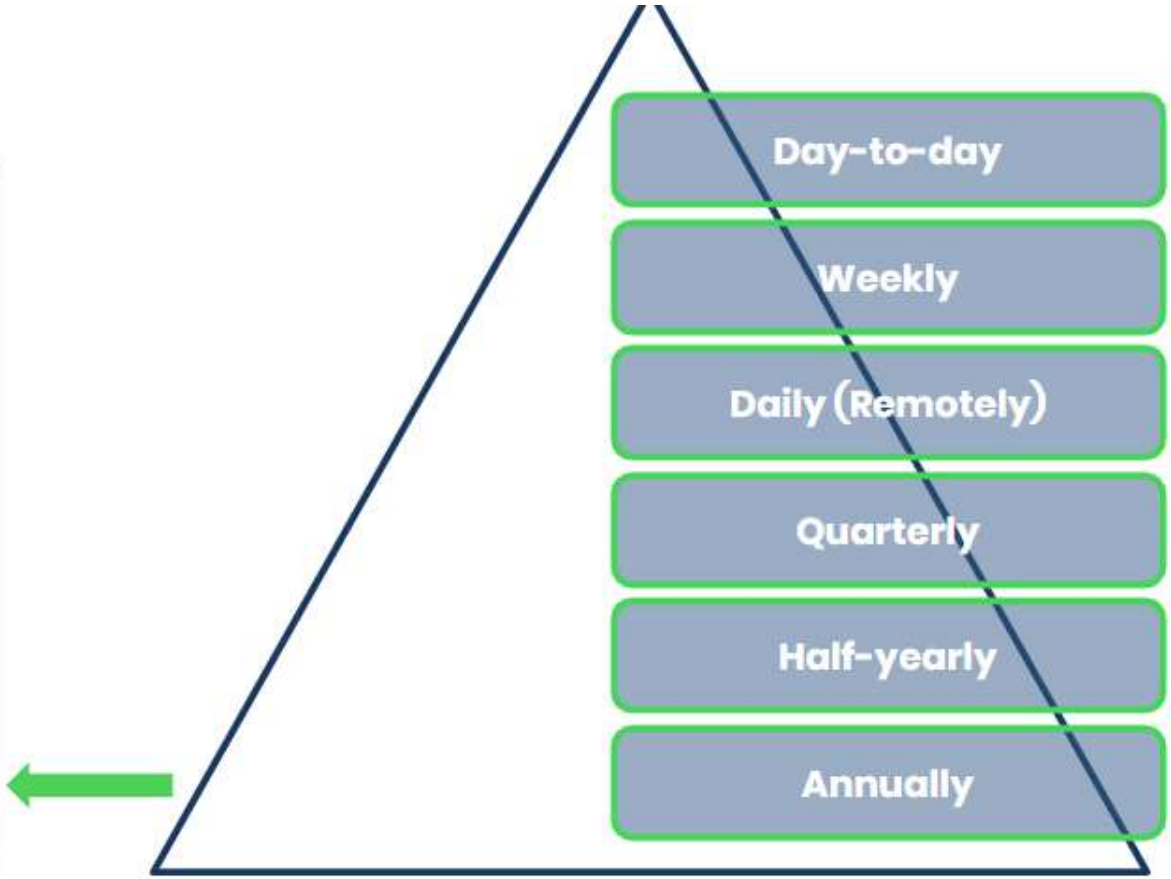


In Half-yearly maintenance, **ensure** all electrical connections are kept clean and tight.



In Annual maintenance:

- Check mechanical integrity of the array structure.
- Check all cabling for mechanical damage.
- Check the output voltage and current of each string of the array and compare to the expected output under the existing conditions.
- Check the operation of the PV array DC isolator.



7.2.2: Maintenance of Solar PV Module

- Regular maintenance of solar PV modules involves
  - cleaning accumulated dust and debris,
  - inspecting for damage, and
  - ensuring proper electrical connections.
- By adhering to a maintenance schedule, you optimize energy production and extend the lifespan of your solar PV system.



Maintenance of Solar PV Module



Solar PV Module Maintenance



DID YOU KNOW?

INTERESTING FACT

Energy production can decrease by up to 20% if modules are not cleaned regularly.

Solar PV Modules: Cleaning

- The proper techniques for inspecting and cleaning modules, ensuring they remain free from dirt and debris.
- Before commencing any cleaning activities, it's imperative to inspect the site for potential hazards



Solar PV Modules: Cleaning

- Cover all electrical equipment beneath the cleaning area with protective covers.

- **Use a gentle approach**

Remove all accumulated dirt using a sponge or brush with clean water at minimal pressure.



Solar PV Modules: Cleaning

- Avoid harsh chemicals or abrasive cleaning methods, as they can damage the modules over time.
- Where dirt, such as bird droppings, proves challenging to remove, utilize a soft-bristled brush for gentle yet effective cleaning.



Solar PV Modules: Cleaning

Quality of the cleaning water

- Prolonged use of hard water can lead to mineral deposits on the modules, affecting their efficiency.
- Consider conditioning the cleaning water through a water softener.
- Water with a hardness level below 200 ppm to maintain module integrity.



Solar PV Modules: Cleaning





- Clean modules, either in the early morning or late evening to avoid thermal shock to the glass.
- Ensure that modules are not in direct sunlight during the cleaning process.

Solar PV Modules: Physical Damage

- Inspection for physical damage is essential for the longevity of solar PV systems.
- Annually, inspect the array for broken modules



Solar PV Modules: Physical Damage





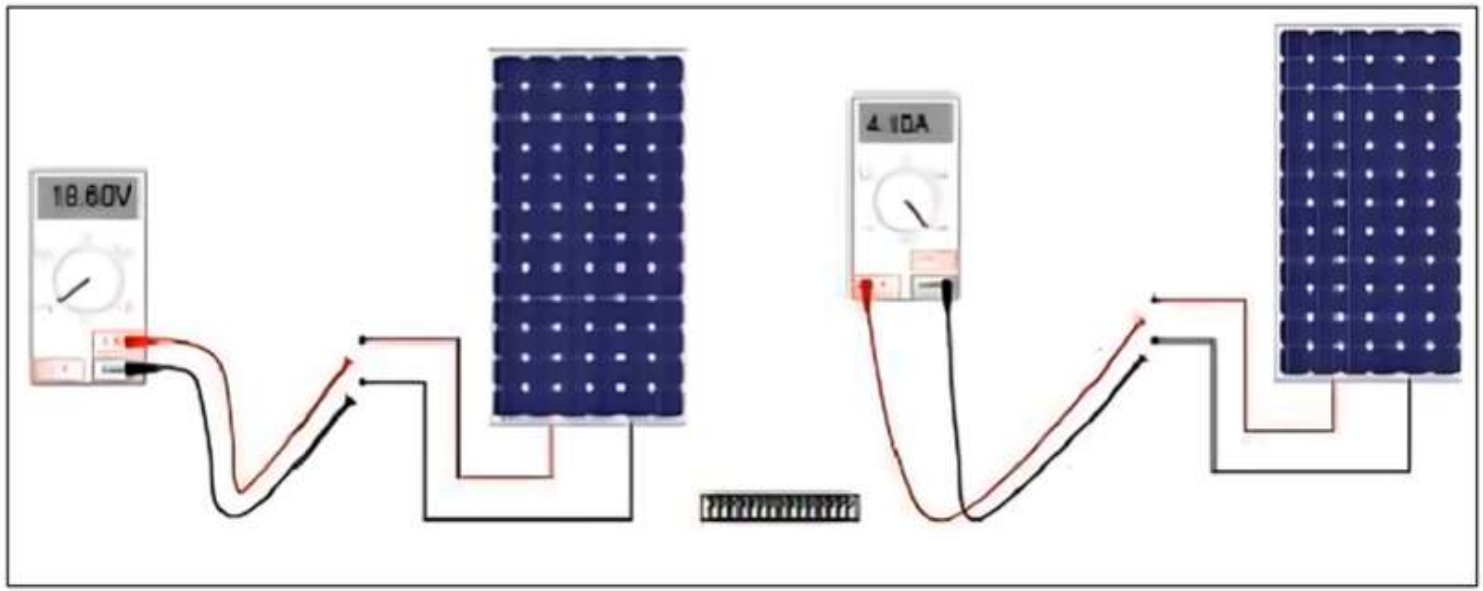
- Promptly report it to the supervisor or engineer for replacement.
- Follow proper isolation procedures before replacing any broken modules to ensure safety.

Solar PV Modules: Performance Monitoring



- Monitoring the performance of PV modules is a key to identify potential issues.

Solar PV Modules: Performance Monitoring



- Regularly check the output voltage and current of each string of the array, comparing it to expected values under existing condition

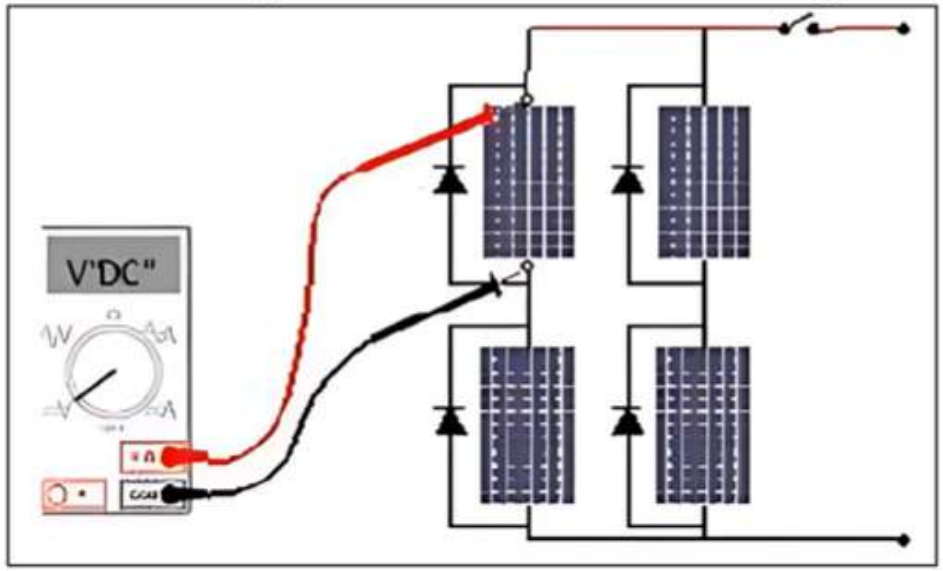
Solar PV Modules: Performance Monitoring

- Efficient utilization of tools like a DC clamp-on ammeter to determine the array's output current during sunny weather



Solar PV Modules: Inspection and Maintenance

Measure the Open Circuit Voltage Voltage

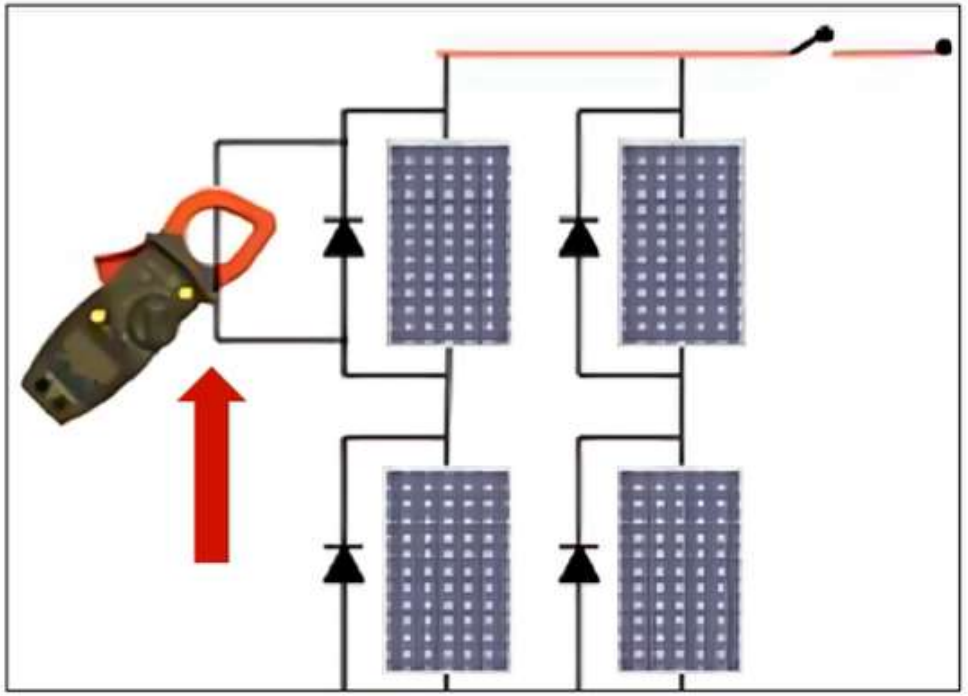


Comparing the measured amount of open circuit voltage from the array against the manufacturer's specifications

Solar PV Modules: Inspection and Maintenance

Short Circuit Current

- Connect the DC meter leads to the positive and negative terminals of each module and set the meter to the 10A range



Solar PV Modules: Inspection and Maintenance

- Regularly check and tighten conduit connections.
- Inspect the area around the array for signs of animal damage and consider investing in monitoring systems for optimal performance



7.2.2: Maintenance of Solar PV Module

Monitor Power Generation

- In a grid-connected system, monitoring power generation and export to the grid is essential for ensuring optimal performance and efficient energy utilization.

- This is typically achieved by observing and recording daily energy generation reports, which are automatically sent by the inverter communication system.



7.2.2: Maintenance of Solar PV Module

Monitor Power Generation

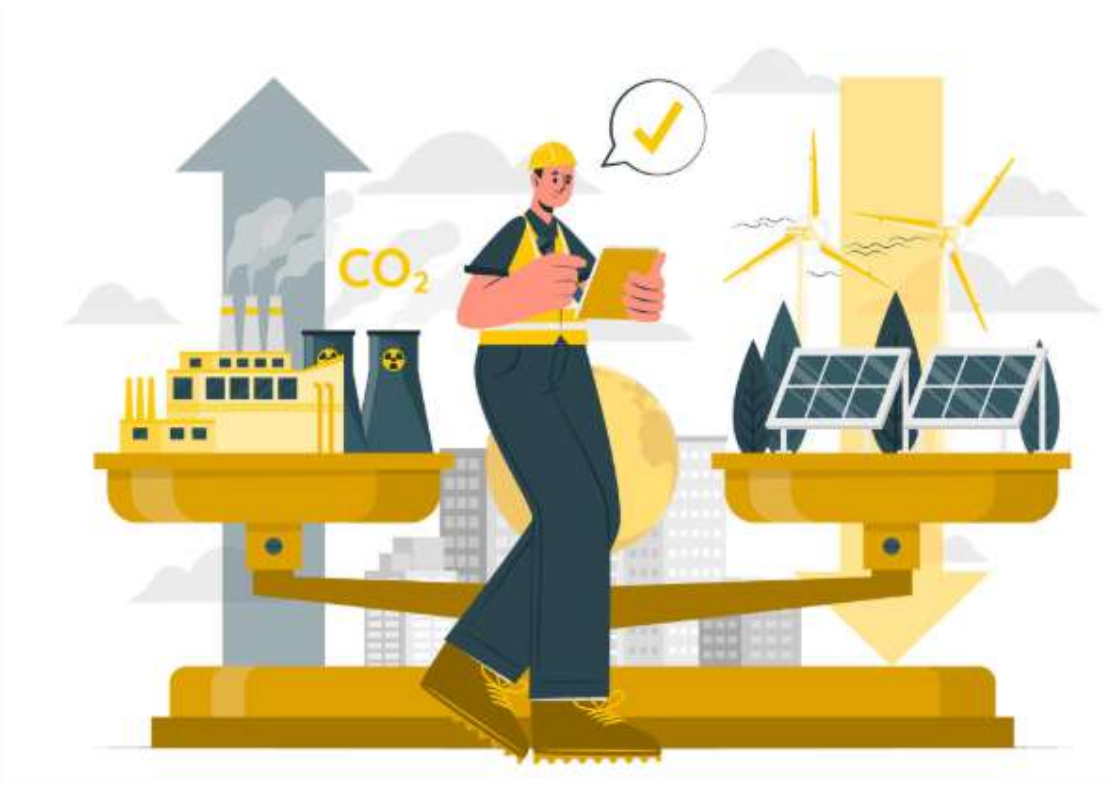
- However, if there is any issues with the inverter communication system, it's imperative to take immediate action.

- In such cases, promptly report the malfunction to the supervisor or engineer responsible for maintenance.



Monitor Power Generation

By staying vigilant and proactive in monitoring power generation, we can ensure the smooth operation of the system and maximize its contribution to sustainable energy production.



7.2.2: Maintenance of Solar PV Module

- Maintenance of solar PV system components involves
  - regular inspection and cleaning of solar panels,
  - checking connections, and
  - monitoring the performance of inverters, batteries, cables, and switches.

- By addressing any issues promptly and adhering to a maintenance schedule, you can maximize energy production and ensure the longevity of your solar PV system.



Maintenance of Solar PV Components



Battery Maintenance Guidelines

Ensure that batteries are clean and topped up with electrolyte.

For Weekly Maintenance:

- Ensure the battery terminals, surfaces, and surrounding areas are clean and free from dirt or corrosion.
- Monitor the battery state of charge (SOC) using a hydrometer.



For Monthly Maintenance:

- Check the electrolyte level and top up if necessary.
- Wipe any electrolyte residue from the top of the battery.







- Inspect all terminals for corrosion and loose cable connections.
- Clean and tighten connections as needed.
- Apply anti-oxidant to exposed wires and terminals after cleaning.





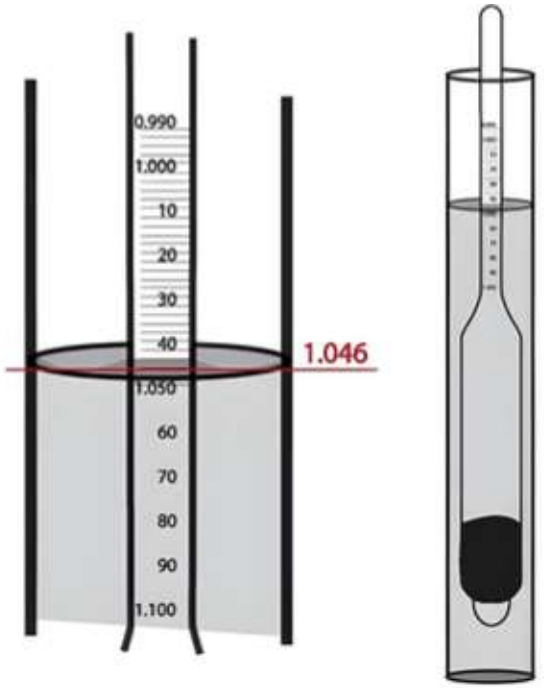




- Check for any newly added loads and ensure the system is not overloaded.

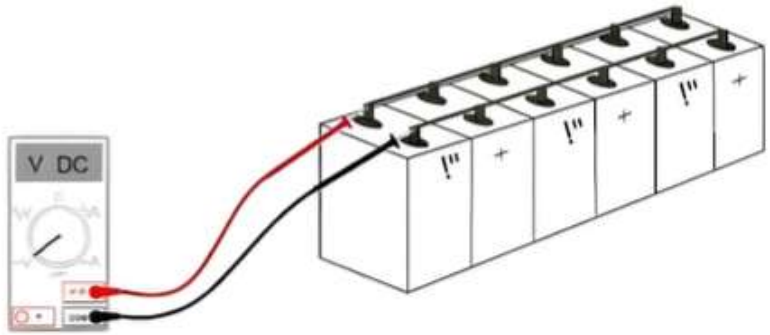
For State of Charge Check:

- Use a hydrometer to measure the specific gravity of the electrolyte, typically ranging between 1.120 and 1.265.
- A reading of 1.120 indicates a fully discharged battery, while 1.265 indicates a fully charged battery.



- Perform an actual load test by operating system loads from the batteries for five minutes.

- Then, disconnect the batteries from the rest of the system and measure the voltage across the terminals of each battery using a DC voltmeter or multi-meter.





By following these maintenance procedures regularly, you can ensure the longevity and efficiency of batteries, optimizing the performance of solar PV system.

Inverter Maintenance Guidelines





- Inspect and clean the inverter, ensuring removal of any dust or dirt accumulation and checking system wiring for secure connections.

- Verify the inverter's operation during inspection and record its current draw in both idle and operational states.







- Thoroughly examine all inverter wiring for signs of damage or deterioration, including loose, broken, corroded, or burnt connections.

- Look out for potential short circuits or ground faults in the wiring.

- Check for any obstructions blocking ventilation in the inverter room, hindering free airflow necessary for natural cooling, and promptly remove them.





Inverter Maintenance Procedure

A. Power Off the Inverter

WARNING:

- Always power off the inverter before performing any maintenance to avoid damage or electric shocks.
- Allow components to discharge after power-off

➤ **Step 1 :** Send a shutdown command to the inverter.

➤ **Step 2:** Turn off the AC switch between the inverter and the utility grid.

➤ **Step 3:** Turn off the DC switch of the inverter.







B. Removing the Inverter

WARNING:

- Ensure the inverter is powered off
- Wear proper personal protective equipment (PPE) before any operations.

➤ **Step 1:** Disconnect all cables, including DC, AC, communication, and PE cables

➤ **Step 2:** Remove the inverter from the mounting plate by grabbing handles or using a hoist.





B. Removing the Inverter

WARNING:

- Ensure the inverter is powered off
- Wear proper personal protective equipment (PPE) before any operations.

➤ **Step 3:** Remove the mounting plate.

➤ **Step 4:** Store the inverter properly, ensuring storage conditions meet requirements for future use.



C. Disposing of the Inverter

- If the inverter is no longer operational, dispose of it according to local regulations for electrical equipment waste.

- Never dispose of the inverter with household waste.



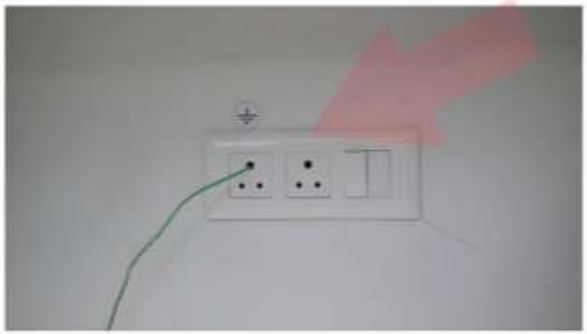


By adhering to these maintenance procedures, you ensure the longevity and efficiency of the inverters, contributing to the smooth operation of solar PV system.

Cables, Connectors, and Switches Maintenance







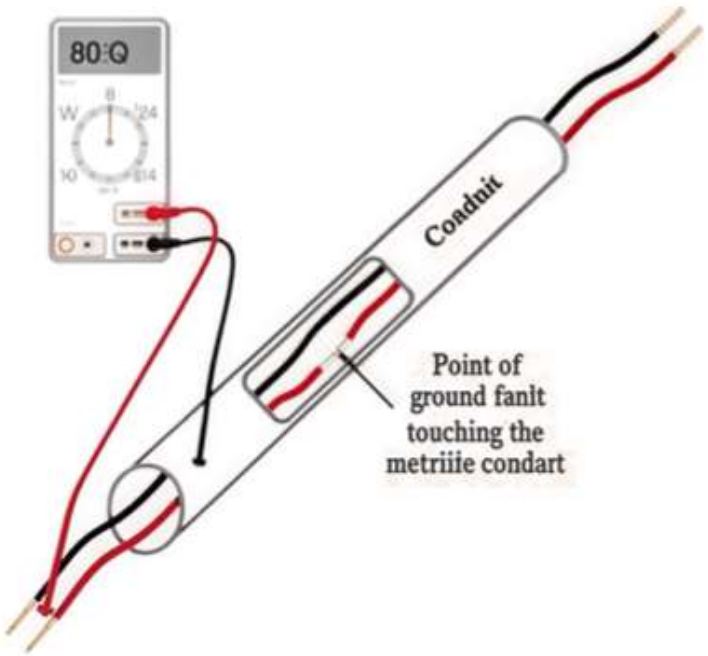
- Visually inspected all conduit and wire insulation for damage, ensuring integrity.
- Checked for any loose, broken, corroded, or burnt wiring connections, rectifying as needed.
- Verified proper wiring and conduit connections for all equipment.
- Secured all wiring by gently but firmly testing connections for stability.



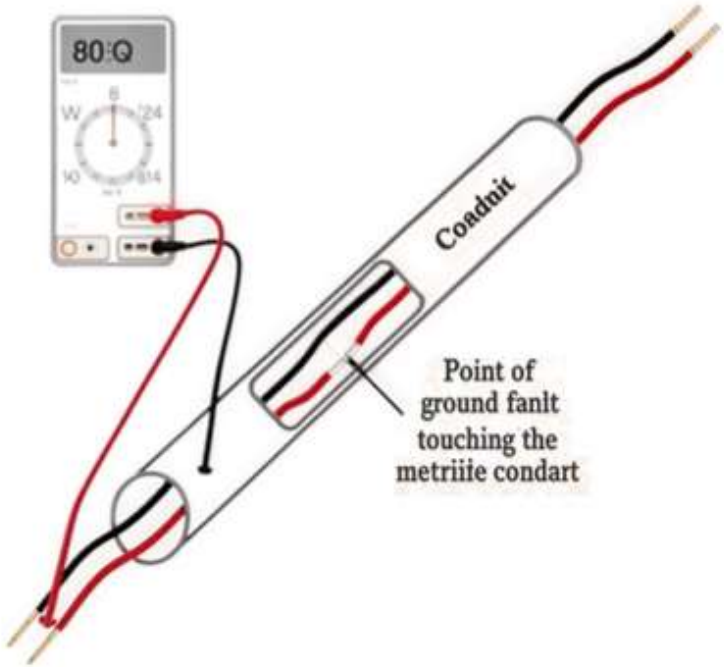
- Thoroughly examined terminals and wires for any signs of damage or deterioration.
- Tested the operation of PV array DC isolators to ensure functionality.
- Inspected array wiring for physical damage and wind-induced wear.
- Assessed array mounting hardware for tightness and corrosion, addressing any issues promptly.



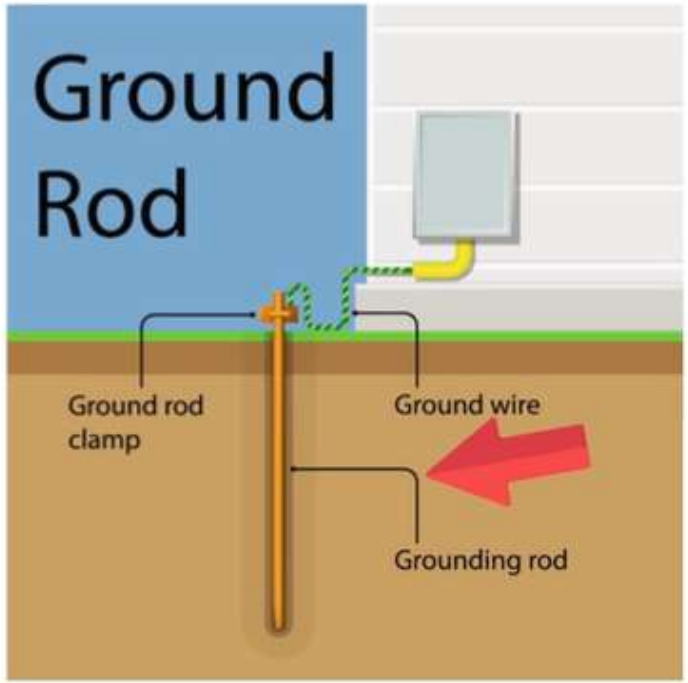
Earthing and Lightning Protection Inspection



- Utilize an ohmmeter or multimeter to confirm continuity across the grounding system.
- Ensure all module frames, metal conduit, connectors, junction boxes, and electrical components are effectively earth grounded.



- By conducting these meticulous inspections and maintenance procedures, Arya guarantees the safety and efficiency of the solar PV system, safeguarding against potential electrical hazards and optimizing its performance.



Unit 7.3: Troubleshooting and Maintenance of PV System

7.3.1: Troubleshooting and Maintenance

Troubleshooting and maintenance of solar PV systems involve identifying and addressing issues such as reduced energy production, faulty connections, or damaged components.

Regular inspections, testing, and prompt repairs ensure optimal system performance and longevity, contributing to sustained renewable energy generation.



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Guidelines for Troubleshooting and Maintenance







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Guidelines for Troubleshooting and Maintenance

Ever wonder what happens when the trusty solar panels go silent on a bright, sunny day?

Don't be left in the dark!



This comprehensive guide empowers you to troubleshoot common solar panel system issues and their solutions, ensuring the clean energy flows smoothly.

Troubleshooting with Solar Photovoltaic System

Problem	Check	Solutions
Load does not operate at all	Switches are turned off or in the wrong position	Put all switches in the correct position
	System circuit breakers or fuses are blown	Reset circuit breaker or replace fuse
	Load is too large for the system	Reduce load size or operating time
	Shadow on solar array	Remove shadow
	Weather is cloudy	Wait until weather is sunny and batteries get charged
	Load itself is defective	Repair or replace load
Load operates poorly or not at all	Voltage drop in the system due to small and long wire	Use bigger wire
	System is overloaded	Reduce load size or operating time
	Ground fault or a defective diode	Correct ground faults or defective diodes
	Check if wiring or connections are loose, broken, burned, or corroded	Repair or replace damaged wiring or connections
	Short-Circuit	Repair short circuits
	Small "Phantom" load keeps inverter idling, draining battery	Turn-off phantom load or replace it with the one not requiring PV power
	Wiring polarity is reversed	Correct the wiring polarity

Troubleshooting with Batteries

Problem	Check	Solutions
Battery voltage remains low	Battery is too cold	Insulate battery enclosure
	Shadow on solar panels	Remove shadow
	Load operates too long	Reduce operating time
	Cloudy weather for several days	Wait till weather is sunny and restrict use of load
	Defective load	Repair or replace load
	Load is too large	Reduce load size
Battery do not accept charge Low electrolyte level	Load is too large	Reduce load size
	Load operates too long	Reduce operating time
	Shadow on solar panels	Remove shadow
	Cloudy weather for several days	Wait till weather is sunny and restrict use of load
Electrolyte leakage	Batteries are overcharged	Add distilled water
	Battery container is broken or leaking	Replace battery and report to dealer/manufacturer
High water loss due to overcharging Voltage loss overnight	Batteries are overcharged	Repair or replace charge controller
	Blocking diode is faulty	Replace diode
When battery voltage constantly remains high	Charge controller is faulty	Replace charge controller with lower charge termination setting
	Battery capacity is too small for system	Increase battery capacity
	Charge controller is misadjusted	Adjust charge controller

Troubleshooting with Charge Controllers

Problem	Possible Causes	Solutions
Erratic controller operation or loads being disconnected improperly	Timer not synchronized with actual time	Either wait until automatic reset next day or disconnect array wait 10 seconds and reconnect array
	Electrical "noise" from the inverter	Connect inverter directly to batteries, put filters on load
	Low battery voltage	Repair, replace battery
	High surge from load	Use large wire to load or add batteries in parallel
	Otherwise, faulty charge controller, possibly from lightning damage	Repair or replace charge controller and check system grounding
	Adjustable low voltage disconnect set incorrectly	Reset low voltage setting
	Load switch in wrong position on controller	Reset switch to correct position
	Charge controller has low voltage disconnect feature	If necessary, replace charge controller with one with a low voltage disconnect feature
Fuse to array blows	Array short circuited with batteries still connected	Disconnect batteries when testing array's short circuit current
	Current output of array too high for charge controller	Replace charge controller with one with a higher rating
Fuse to load blows	Current draw of load too high for charge controller	Reduce load size or increase charge controller size
	Surge current draw of load too high for charge controller	Reduce load size or increase charge controller size

Troubleshooting with Inverters

Problem	Possible Causes	Solutions
No output from the inverter	Check switch, fuse or circuit breaker open, blown or tripped or wiring broken or corroded	Close switch, replace or reset fuse or circuit breaker or repair wiring or connections
	Check low voltage disconnect on inverter or charge controller open	Allow batteries to recharge
	Check if there is time delay on inverter start-up from idle	Wait a few seconds after starting loads
Loads operating improperly	Excessive current draw by load	Reduce size of loads
	Defective inverter	Replace inverter/report to dealer/manufacturer
Motors operating at wrong speeds	Inverter not equipped with frequency control	Replace inverter with one equipped with frequency control/report to dealer/manufacturer
Inverter circuit breaker trips	Load operating or surge current too high	Reduce size of loads or replace inverter with one of larger capacity/report to dealer/manufacturer
Inverter DC circuit breaker trips	Inverter capacitors not charged up on initial start up	Report to dealer/manufacturer

Troubleshooting with Solar Photovoltaic Arrays

Problem	Possible Causes	Solutions
No current from array	Check switches, fuses or circuit breaker open, blown or tripped or wiring broken or corroded	Close switches, replace fuses, reset circuit breakers, repair or replace damaged wiring
	Check if module is damaged or broken	Report
No voltage from array	Check switches, fuses or circuit breaker open, blown or tripped or wiring broken or corroded	Close switches, replace fuses, reset circuit breakers, repair or replace damaged wiring
	Check if module is damaged or broken	Report

Troubleshooting with Solar Photovoltaic Arrays

Problem	Possible Causes	Solutions
Low array voltage	Some modules shaded	Remove source of shading or relocate the array
	Some array interconnections broken or corroded	Repair interconnections
	Defective bypass or blocking diode	Repair defective diodes
	Some modules damage or defective	Replace affected modules
	Full sun not available	Wait for sunny weather
	Modules are dirty	Clean modules
	Array tilt or orientation incorrect	Correct tilt/orientation
	Check if module is damaged or broken	Report
	Some modules in series disconnected	Reconnect
	Bypass diode defective	Replace diodes
	Wiring from too long or small	Use larger wire
Low array current	Check if full sun is not available	Measure current at noon
	Check if modules are dirty	Clean modules
	Check if array tilt or orientation incorrect	Correct tilt/orientation
	Check if battery is fully charged	Wait till battery is charged
	Check if module is damaged or broken	Report
	Check if array is not giving expected load current at full sun	Report

Guidelines for Troubleshooting and Maintenance



By implementing these maintenance practices, you ensure that the solar PV system continues to shine brightly, harnessing the power of the sun for years to come.